

Assessment tool

CO4-AT2

Course code:csa0905

Course name:java programming

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Java applet

Tools used:jdk 8

AIM:TO PERFORM ARTHMETIC OPERATIONSIN APPLET

Procedure:first create a java file using notepad

Create a html file

Save both in a folder

Run the java filee then class file will be created

Then run the html file

CODE:

Java file

```
import java.applet.Applet;
```

```
import java.awt.Graphics;
```

```
public class ArithmeticApplet extends Applet {
```

```
    public void paint(Graphics g) {
```

```
int a = 20;

int b = 10;


g.drawString("Arithmetic Operations", 50, 50);
g.drawString("Addition: " + (a + b), 50, 80);
g.drawString("Subtraction: " + (a - b), 50, 110);
g.drawString("Multiplication: " + (a * b), 50, 140);
g.drawString("Division: " + (a / b), 50, 170);
}
}
```

Html file:

```
<html>

<head>

    <title>Arithmetic Applet</title>

</head>

<body>


<h2>Arithmetic Operations</h2>


<applet code="ArithmeticApplet.class" width="400" height="250">

</applet>


</body>

</html>
```

Output:

Aim:to demonstrate lifecycle of applete

Code:

Java code

```
public void start() {  
    message = "start() method called";  
    System.out.println("start() called");  
}  
  
public void paint(Graphics g) {  
    g.setColor(Color.BLUE);  
    g.setFont(new Font("Arial", Font.BOLD, 20));  
    g.drawString("Applet Life Cycle", 80, 50);  
  
    g.setColor(Color.BLACK);  
    g.setFont(new Font("Arial", Font.PLAIN, 16));  
    g.drawString(message, 80, 100);  
  
    g.drawString("init() → start() → paint() → stop() → destroy()", 30, 150);  
}
```

Applet Viewer: ArithmeticApplet.class

Applet

Arithmetic Operations

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2

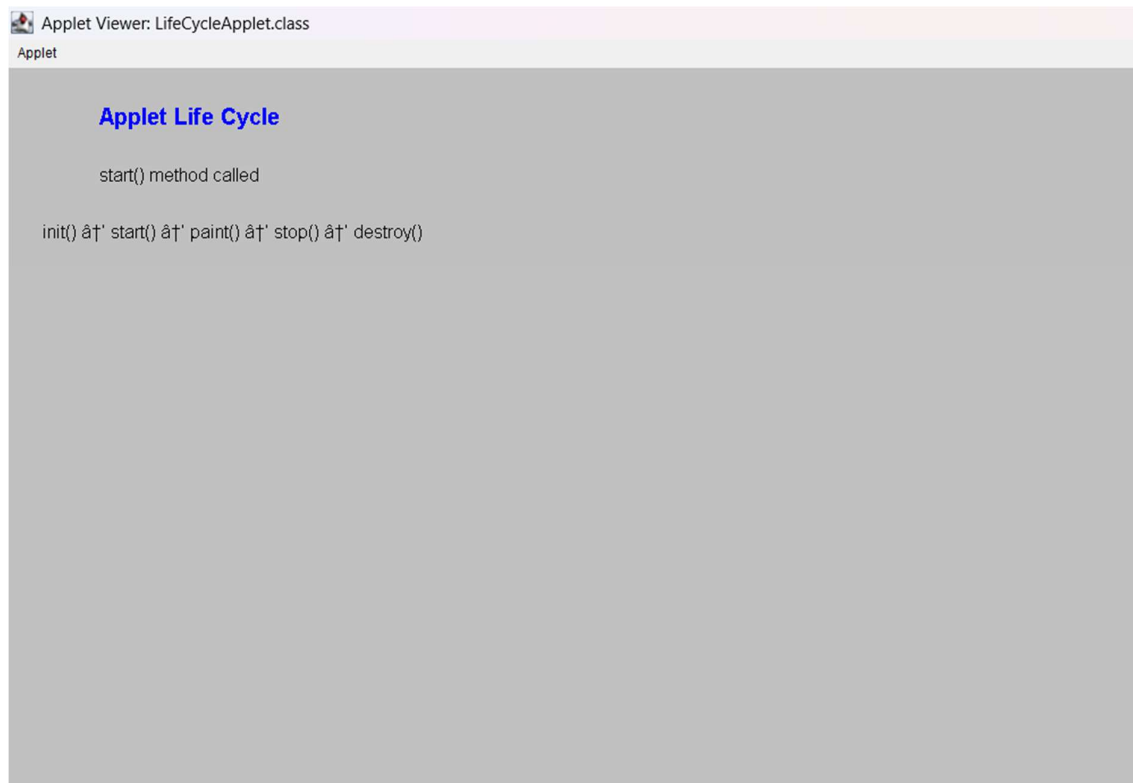
```
public void stop() {  
    System.out.println("stop() called");  
}
```

```
public void destroy() {  
    System.out.println("destroy() called");  
}  
}
```

Html:

```
<html>  
<head>  
    <title>Applet Life Cycle</title>  
</head>  
  
<body>  
  
    <h2>Java Applet Life Cycle</h2>  
  
    <applet code="LifeCycleApplet.class" width="500" height="220">  
    </applet>  
  
</body>  
</html>
```

Output:



Aim:to create different shapes

Code

Java code:

```
import java.applet.Applet;
```

```
import java.awt.*;
```

```
public class ShapesApplet extends Applet {
```

```
    public void paint(Graphics g) {
```

```
        setBackground(Color.WHITE);
```

```
        // Line
```

```
        g.setColor(Color.BLACK);
```

```
g.drawLine(30, 50, 180, 50);
```

```
// Rectangle
```

```
g.setColor(Color.BLUE);
```

```
g.drawRect(30, 80, 120, 70);
```

```
// Filled Rectangle
```

```
g.setColor(Color.CYAN);
```

```
g.fillRect(180, 80, 120, 70);
```

```
// Circle
```

```
g.setColor(Color.RED);
```

```
g.drawOval(30, 180, 100, 100);
```

```
// Filled Circle
```

```
g.setColor(Color.PINK);
```

```
g.fillOval(180, 180, 100, 100);
```

```
// Oval
```

```
g.setColor(Color.GREEN);
```

```
g.drawOval(320, 80, 150, 80);
```

```
// Triangle
```

```
g.setColor(Color.ORANGE);
```

```
int x[] = {350, 300, 400};
```

```
int y[] = {180, 280, 280};
```

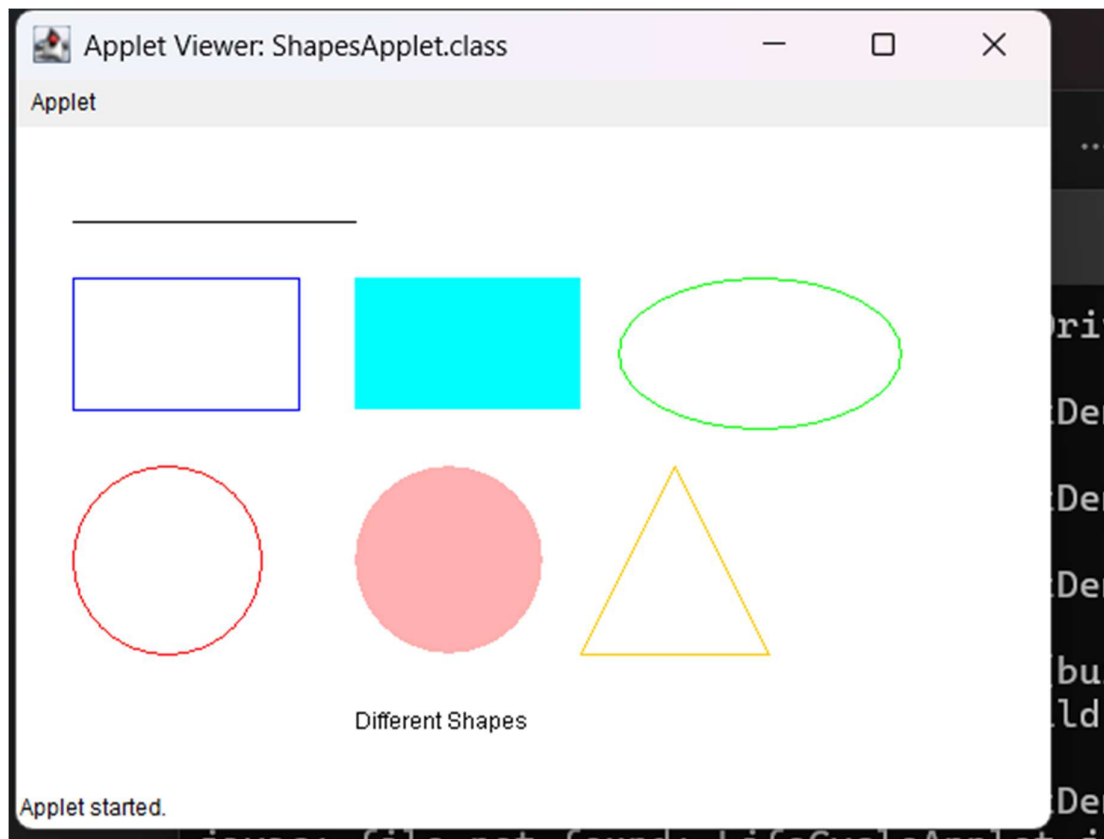
```
g.drawPolygon(x, y, 3);
```

```
// Text  
g.setColor(Color.BLACK);  
g.drawString("Different Shapes", 180, 320);  
}  
}
```

Html code:

```
<html>  
  
<head>  
  <title>Shapes Applet</title>  
</head>  
  
<body>  
  
  <h2>Drawing Different Shapes</h2>  
  
  <applet code="ShapesApplet.class" width="550" height="350">  
    </applet>  
  
</body>  
</html>
```

Output:



4) student application form using various awt components

Code:

```
package org.example;
import java.awt.*;
import java.awt.event.*;
public class StudentApplicationForm extends Frame implements ActionListener {
    TextField name, age;
    TextArea address;
    Checkbox male, female, coding, reading, sports;
    CheckboxGroup gender;
    Choice course;
    Button submit, reset;
    StudentApplicationForm() {
        setTitle("Student Application Form");
        setSize(500, 500);
    }
}
```

```
setLayout(new GridLayout(8, 2));
add(new Label("Name:"));
name = new TextField();
add(name);
add(new Label("Age:"));
age = new TextField();
add(age);
add(new Label("Gender:"));
gender = new CheckboxGroup();
Panel genderPanel = new Panel();
male = new Checkbox("Male", gender, true);
female = new Checkbox("Female", gender, false);
genderPanel.add(male);
genderPanel.add(female);
add(genderPanel);
add(new Label("Course:"));
course = new Choice();
course.add("CSE");
course.add("Data Science");
course.add("IT");
course.add("ECE");
add(course);
add(new Label("Address:"));
address = new TextArea();
add(address);
add(new Label("Hobbies:"));
Panel hobbyPanel = new Panel();
coding = new Checkbox("Coding");
reading = new Checkbox("Reading");
sports = new Checkbox("Sports");
hobbyPanel.add(coding);
hobbyPanel.add(reading);
hobbyPanel.add(sports);
add(hobbyPanel);
submit = new Button("Submit");
reset = new Button("Reset");
submit.addActionListener(this);
reset.addActionListener(this);
add(submit);
add(reset);
addWindowListener(new WindowAdapter() {
```

```

        public void windowClosing(WindowEvent e) {
            dispose();
        }
    });
    setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == submit) {
        System.out.println("Student Application");
        System.out.println("Name: " + name.getText());
        System.out.println("Age: " + age.getText());
        System.out.println("Course: " + course.getSelectedItem());
        System.out.println("Address: " + address.getText());
    } else if (e.getSource() == reset) {
        name.setText("");
        age.setText("");
        address.setText("");
        gender.setSelectedCheckbox(null);
        course.select(0);
        coding.setState(false);
        reading.setState(false);
        sports.setState(false);
    }
}

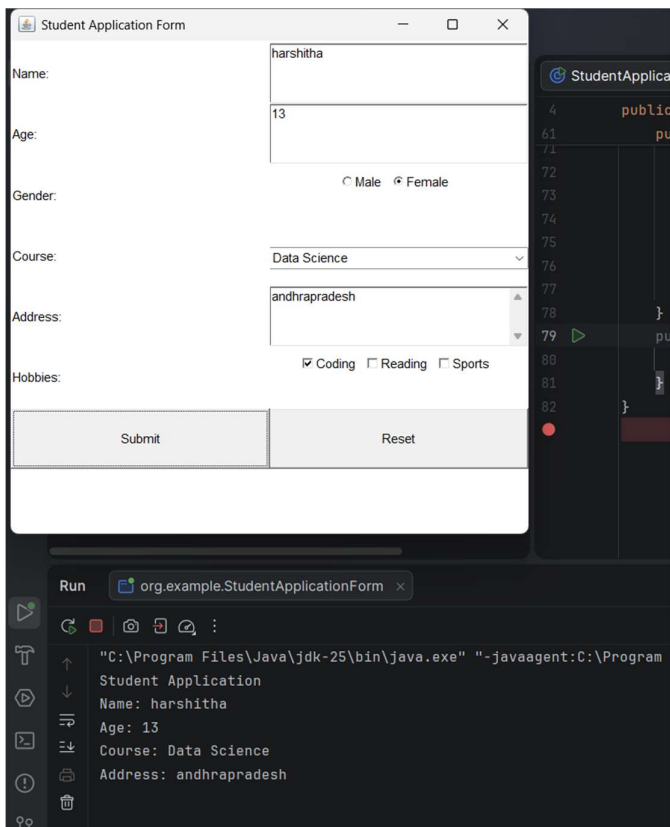
public static void main(String[] args) {
    new StudentApplicationForm();
}
}

```

output:

```
Student Application
Name: harshitha
Age: 18
Course: CSE
Address: chennai

Process finished with exit code 0
```



5)using different layout and performing events

Border layout:

```
package org.example;
import java.awt.*;
import java.awt.event.*;
public class BorderDemo extends Frame implements ActionListener {
    Label label;
    Button button;
    BorderDemo() {
        setTitle("BorderLayout Demo");
        setSize(400, 300);
        setLayout(new BorderLayout(10, 10));
        label = new Label("Click a button", Label.CENTER);
        add(label, BorderLayout.CENTER);
        button = new Button("Click Me");
        button.addActionListener(this);
        add(button, BorderLayout.SOUTH);
        add(new Button("North"), BorderLayout.NORTH);
        add(new Button("East"), BorderLayout.EAST);
        add(new Button("West"), BorderLayout.WEST);

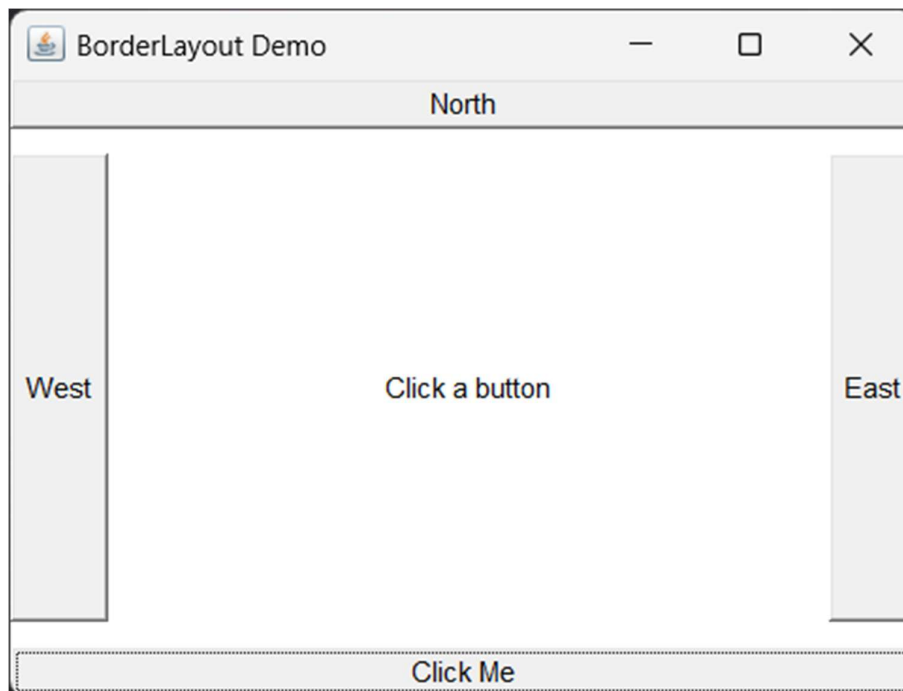
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        label.setText("Button Clicked!");
    }

    public static void main(String[] args) {
        new BorderDemo();
    }
}
```

output:



Grid layout:

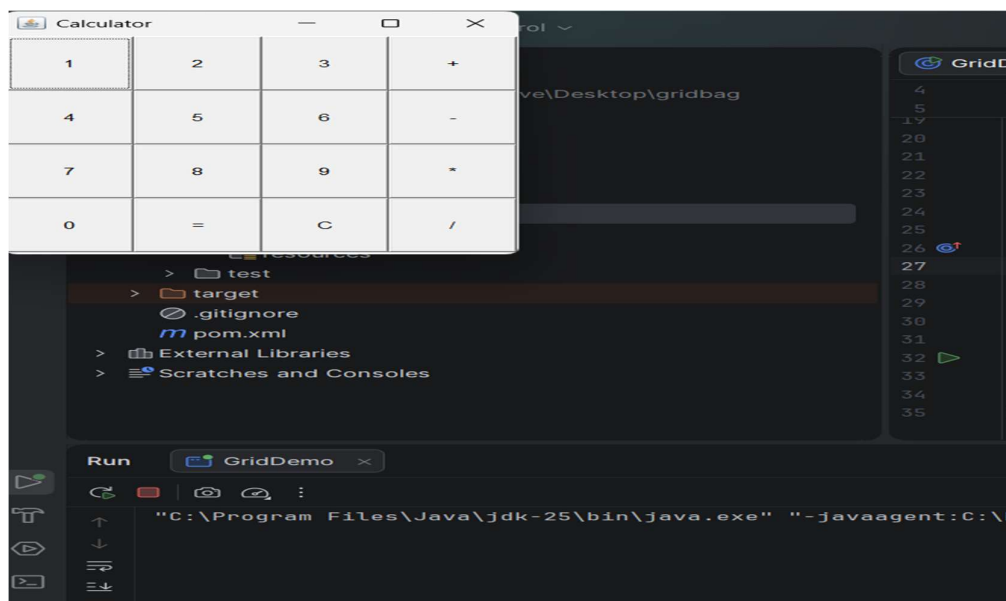
```
package org.example;
import java.awt.*;
import java.awt.event.*;
public class GridDemo extends Frame {
    GridDemo() {
        setTitle("Calculator");
        setSize(300,300);
        setLayout(new GridLayout(4,4));
        add(new Button("1"));
        add(new Button("2"));
        add(new Button("3"));
        add(new Button("+"));
        add(new Button("4"));
        add(new Button("5"));
        add(new Button("6"));
        add(new Button("-"));
        add(new Button("7"));
        add(new Button("8"));
        add(new Button("9"));
        add(new Button("*"));
        add(new Button("0"));
    }
}
```

```

add(new Button("="));
add(new Button("C"));
add(new Button("/"));
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});
setVisible(true);
}
public static void main(String[] args) {
    new GridDemo();
}
}

```

output:



Card layout:

Code:

```

package org.example;

import java.awt.*;
import java.awt.event.*;

```

```

public class CardDemo extends Frame implements ActionListener {
    CardLayout card;
    Panel p;
    TextField name;

    CardDemo() {
        setTitle("Card Layout");
        setSize(400,300);

        card=new CardLayout();
        p=new Panel(card);

        Panel p1=new Panel();
        p1.add(new Label("Enter Name:"));
        name=new TextField(15);
        p1.add(name);

        Button b=new Button("Login");
        p1.add(b);
        b.addActionListener(this);

        Panel p2=new Panel();
        p2.add(new Label("Welcome to the page!"));
        p2.add(new Button("Login Successful"));

        p.add(p1,"login");
        p.add(p2,"welcome");

        add(p);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                System.exit(0);
            }
        });

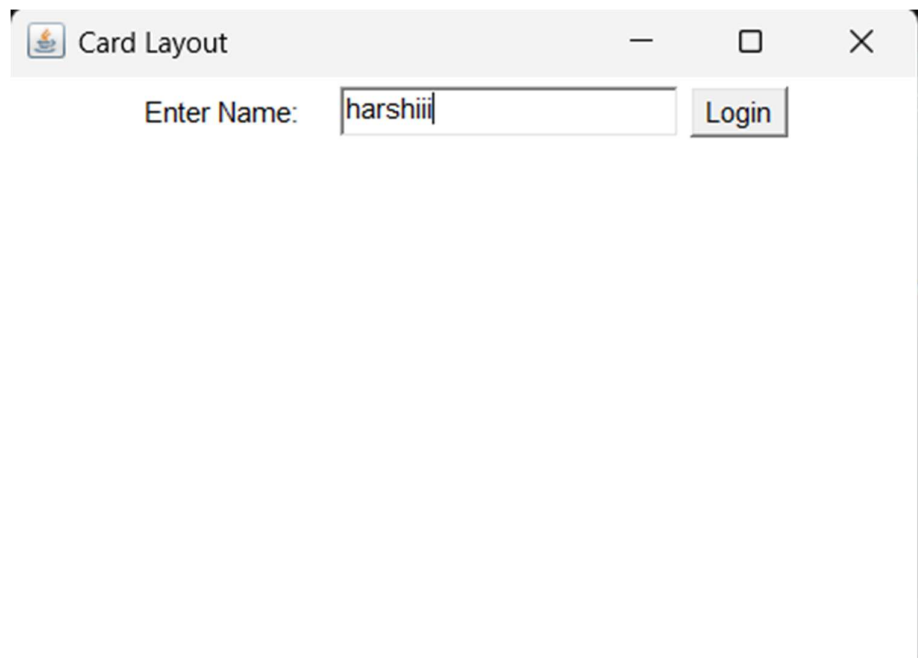
        setVisible(true);
    }

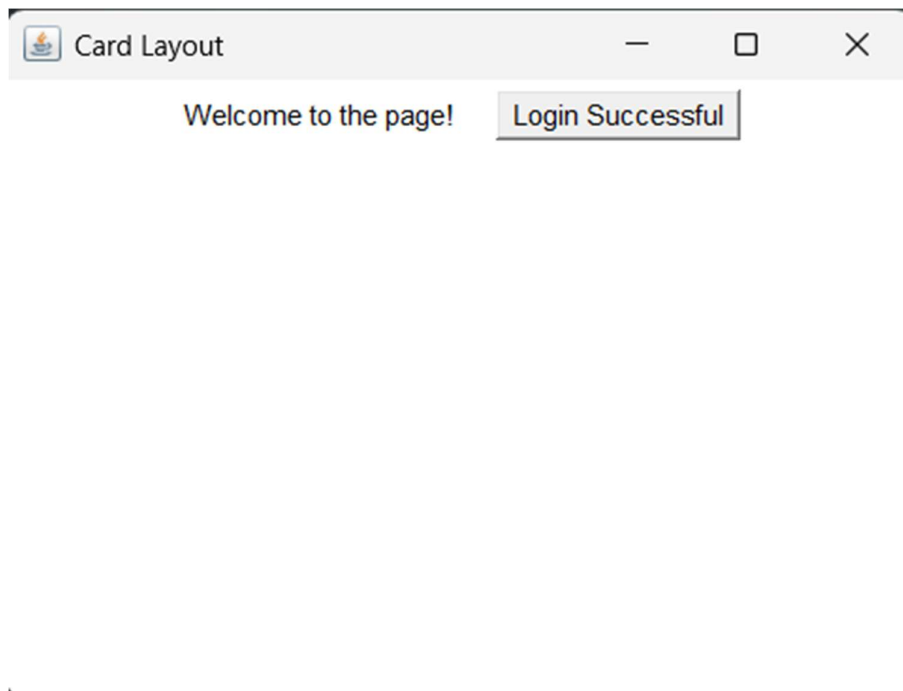
    public void actionPerformed(ActionEvent e) {

```

```
        card.show(p,"welcome");  
    }  
  
    public static void main(String[] args) {  
        new CardDemo();  
    }  
}
```

Output:





Gridbag layout:

Code:

```
package org.example;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class GridBagDemo extends Frame {
```

```
    TextField name, roll, course;
```

```
    GridBagDemo() {
```

```
        setTitle("Student Registration");
```

```
        setSize(400,300);
```

```
        setLayout(new GridBagLayout());
```

```
GridBagConstraints g = new GridBagConstraints();
```

```
g.gridx=0;
```

```
g.gridy=0;
```

```
add(new Label("Name"),g);
```

```
g.gridx=1;
```

```
name=new TextField(15);
```

```
add(name,g);
```

```
g.gridx=0;
```

```
g.gridy=1;
```

```
add(new Label("Roll No"),g);
```

```
g.gridx=1;
```

```
roll=new TextField(15);
```

```
add(roll,g);
```

```
g.gridx=0;
```

```
g.gridy=2;
```

```
add(new Label("Course"),g);
```

```
g.gridx=1;
```

```
course=new TextField(15);
```

```
add(course,g);
```

```
g.gridx=1;
```

```
g.gridy=3;
```

```

Button b=new Button("Submit");
add(b,g);

b.addActionListener(e -> {
    System.out.println("Student Details");
    System.out.println("Name: "+name.getText());
    System.out.println("Roll No: "+roll.getText());
    System.out.println("Course: "+course.getText());
});

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});

setVisible(true);
}

public static void main(String[] args) {
    new GridBagDemo();
}
}

```

Output:

